



BRAC University

Department of Mathematics and Natural Sciences

LECTURE ON

Real Analysis II (MAT321)

Subspaces

Interior, Exterior, Boundary and Subspaces

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CONDUCTED BY

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Interior, Exterior, Frontier and Boundary Points

Interior Point

Let A be a subset of a metric space (X, d) . A point $a \in A$ is called an *interior point* of A if there exists $r > 0$ such that

$$a \in S_r(a) \subseteq A.$$

Interior of a Set

The set of all interior points of A is called the *interior* of A and is denoted by $\text{int } A$, where

$$\text{int } A = \{x \in A : S_r(x) \subseteq A \text{ for some } r > 0\}.$$

Clearly, $\text{int } A$ is an open subset of A .

Exterior Point

A point $x \in X$ is called an *exterior point* of A if it is an interior point of the complement of A , that is, if there exists $r > 0$ such that

$$S_r(x) \subseteq A^c \quad \text{or equivalently} \quad S_r(x) \cap A = \emptyset.$$

Exterior of a Set

The set of all exterior points of A , denoted by $\text{ext } A$, is called the *exterior* of A . By definition,

$$\text{ext } A = \text{int } A^c.$$

Hence $\text{ext } A$ is open and is the largest open set contained in A^c . Moreover,

$$\text{ext } A = (\overline{A})^c.$$

Boundary Point

A point $x \in X$ is called a *boundary point* of A if every open sphere centered at x intersects both A and its complement A^c , that is, for every $r > 0$,

$$S_r(x) \cap A \neq \emptyset \quad \text{and} \quad S_r(x) \cap A^c \neq \emptyset.$$

Boundary of a Set

The set of all boundary points of A is called the *boundary* of A and is denoted by ∂A , where

$$\partial A = \overline{A} \setminus \text{int } A.$$

Equivalently,

$$\partial A = \overline{A} \cap \overline{A^c}.$$

② Illustrations of Interior, Exterior, Boundary

Compute the interior, exterior, and boundary of the following subsets.

1. Let $X = \mathbb{R}$ with the usual metric and let $A = \mathbb{Q}$. Then

$$\text{int } A = \emptyset, \quad \text{ext } A = \emptyset, \quad \partial A = \mathbb{R}.$$

2. Let $X = \mathbb{R}$ with the usual metric and let $A = (1, 2) \cup (3, 4)$. Then

$$\text{int } A = A,$$

$$\text{ext } A = (-\infty, 1) \cup (2, 3) \cup (4, \infty),$$

$$\partial A = \{1, 2, 3, 4\}.$$

3. If (X, d) is a discrete metric space and $A \subseteq X$, then

$$\text{int } A = A, \quad \text{ext } A = A^c, \quad \partial A = \emptyset.$$

4. Let $X = \mathbb{N}$ with the usual metric and let $A = \{1, 2, 3\}$. Then

$$\text{int } A = \emptyset, \quad \text{ext } A = \emptyset, \quad \partial A = \mathbb{N}.$$

Properties of Interior

Let A and B be any two subsets of a metric space (X, d) . Then:



1. $\text{int } A$ is the largest open set contained in A , that is,

$$\text{int } A = \bigcup \{G : G \text{ is open and } G \subseteq A\};$$

2. A is open if and only if $A = \text{int } A$;
3. $A \subseteq B$ implies $\text{int } A \subseteq \text{int } B$;
4. $\text{int}(A \cap B) = \text{int } A \cap \text{int } B$;
5. $\text{int}(A \cup B) \supseteq \text{int } A \cup \text{int } B$.

Properties of Exterior of a Set

Let (X, d) be a metric space and let A, B be any two subsets of X .
Then:

1. $\text{ext } A$ is the largest open set contained in A^c ;
2. A is closed if and only if $A^c = \text{ext } A$;
3. $A \subseteq B$ implies $\text{ext } B \subseteq \text{ext } A$;
4. $\text{ext}(A \cap B) \supseteq \text{ext } A \cup \text{ext } B$;
5. $\text{ext}(A \cup B) = \text{ext } A \cap \text{ext } B$.

💡 Properties of Boundary

Let (X, d) be a metric space, and let A, B be subsets of X . Then:

1. $\partial A = \overline{A} \cap \overline{A^c} = \overline{A} \setminus \text{int } A$;
2. $\partial A = \emptyset$ if and only if A is both open and closed;
3. A is closed if and only if $\partial A \subseteq A$;
4. A is open if and only if $\partial A \subseteq A^c$;
5. $\partial(A \cap B) \subseteq \partial A \cup \partial B$. The equality holds if $\overline{A} \cap \overline{B} = \emptyset$;
6. $\partial(\text{int } A) \subseteq \partial A$.

Subspaces

Subspace and Induced Metric

Let (X, d) be a metric space and let Y be a non-empty subset of X . The restriction of the metric d to $Y \times Y$, denoted by d_Y , is a metric on Y called the *induced metric*. The metric space (Y, d_Y) is called a *subspace* of (X, d) .

Examples of Subspaces

Verify that:

1. the closed unit interval $[0, 1]$ and the set of all rational numbers $\mathbb{Q}$ are subspaces of $\mathbb{R}$ with the usual metric;
2. the unit circle, the closed unit disc, and the open unit disc are subspaces of the metric space $(\mathbb{C}, d)$ of complex numbers;
3. the real line $\mathbb{R}$ itself is a subspace of the complex plane $\mathbb{C}$.

Open Sphere in a Subspace

Let (X, d) be a metric space, let $Y \subseteq X$, and let $y \in Y$. The open sphere in the subspace (Y, d_Y) centered at y with radius $r > 0$ is denoted by $S_r^Y(y)$ and is defined by

$$S_r^Y(y) = \{x \in Y : d_Y(x, y) < r\}.$$

Relation Between Open Spheres

Show that for every $y \in Y$ and every $r > 0$,

$$S_r^Y(y) = S_r(y) \cap Y,$$

where $S_r(y)$ denotes the open sphere in (X, d) .

💡 Open Sets in Subspaces

Let (X, d) be a metric space and let $Y \subseteq X$. A subset A of Y is open in (Y, d_Y) if and only if there exists a set G open in (X, d) such that

$$A = G \cap Y.$$

🔗 Open Sets in a Subspace vs Ambient Space

Show that if $Y \subseteq X$ and a subset of Y is open in (X, d) , then it is also open in the subspace (Y, d_Y) . Show by examples that the converse need not be true.

❓ Open in a Subspace may Not be open in Superspace

Let $X = \mathbb{R}$ with the usual metric and let

$$Y = [0, 1].$$

Then the open sphere in (Y, d_Y) centered at 0 with radius $\frac{1}{2}$ is

$$S_{1/2}^Y(0) = [0, \frac{1}{2}),$$

which is open in Y but not open in $\mathbb{R}$. Note that Y itself is not open in $\mathbb{R}$.

❓ Example: Real Line vs Complex Plane

Show that an open interval of the real line is not an open subset of the complex plane $\mathbb{C}$ with the usual metric.

Thank You!

We'd love your questions and feedback.

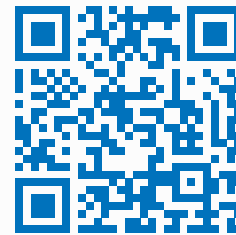
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(Lectures, walkthroughs, and course updates)



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References

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- [2] Walter Rudin, *Principles of Mathematical Analysis*, 3rd Edition, McGraw-Hill, 1976.